

Suite Hard • Simulation Module 03

22 questions • 35 minutes • official SAT Math Hard

Exclude-active: these items are not on current Bluebook full-length tests.

All items are Hard. This is not an adaptive Module 1. It is harder than test day.

Calculator / Desmos is allowed on every item, same as Bluebook.

Question 1 is the first page after this cover. Write answers on the last page.

Read the prompt carefully. Write the job on scratch before you copy numbers.

Domain mix in this module:

Algebra: 5

Advanced Math: 8

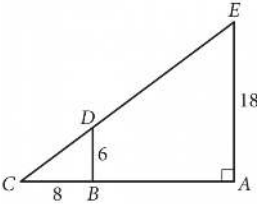
PSDA: 4

Geometry: 5

Do not open the next module until this one is timed and scored.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In the figure above, $\overline{BD}$ is parallel to $\overline{AE}$. What is the length of $\overline{CE}$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$r(t) = 63t - 2t^2$

The function r is defined by the given equation. The function s is defined by $s(t) = r(t) + 1$. Which expression represents the maximum value of $s(t)$?

Answer

- A. $1 + \left(\frac{63}{2}\right)^2$
- B. $1 + 2\left(\frac{63}{4}\right)^2$
- C. $1 - 2\left(\frac{63}{4}\right)^2$
- D. $1 - \left(\frac{63}{2}\right)^2$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

	Site A	Site B	Total
Tulip	35	15	50
Daffodil	31	21	52
Total	66	36	102

The table shows the distribution of two types of flowers at two different sites. If a flower represented in the table is selected at random, what is the probability of selecting a flower from site A, given that the flower is a tulip? (Express your answer as a decimal or fraction, not as a percent.)

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

x	y
k	13
$k + 7$	-15

The table gives the coordinates of two points on a line in the xy -plane. The y -intercept of the line is $(k - 5, b)$, where k and b are constants. What is the value of b ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

Answer

- A. 18
- B. 27
- C. 36
- D. 81

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

When the quadratic function f is graphed in the xy -plane, where $y = f(x)$, its vertex is $(-3, 6)$. One of the x -intercepts of this graph is $(-\frac{17}{4}, 0)$. What is the other x -intercept of the graph?

Answer

- A. $(-\frac{29}{4}, 0)$
- B. $(-\frac{7}{4}, 0)$
- C. $(\frac{5}{4}, 0)$
- D. $(\frac{17}{4}, 0)$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

To study the moisture content in a group of trees, samples from the trunk of each tree were taken from **25** trees and cut in the shape of a cube. The length of the edge of one of these cubes is **2.00** centimeters. If this cube has a mass of **2.56** grams, what is the density of this cube, in grams per cubic centimeter?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

The cost of renting a backhoe for up to **10** days is **\$270** for the first day and **\$135** for each additional day. Which of the following equations gives the cost ***y***, in dollars, of renting the backhoe for ***x*** days, where ***x*** is a positive integer and ***x* ≤ 10**?

Answer

- A. ***y* = 270*x* − 135**
- B. ***y* = 270*x* + 135**
- C. ***y* = 135*x* + 270**
- D. ***y* = 135*x* + 135**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Points A and B lie on a circle with radius 1, and arc $\overline{AB}$ has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc $\overline{AB}$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

Answer

- A. $\frac{1}{(x+1)(4x-5)}$
- B. $\frac{3}{3x-6}$
- C. $-\frac{1}{(x+1)(4x-5)}$
- D. $\frac{9}{(x+1)(4x-5)}$

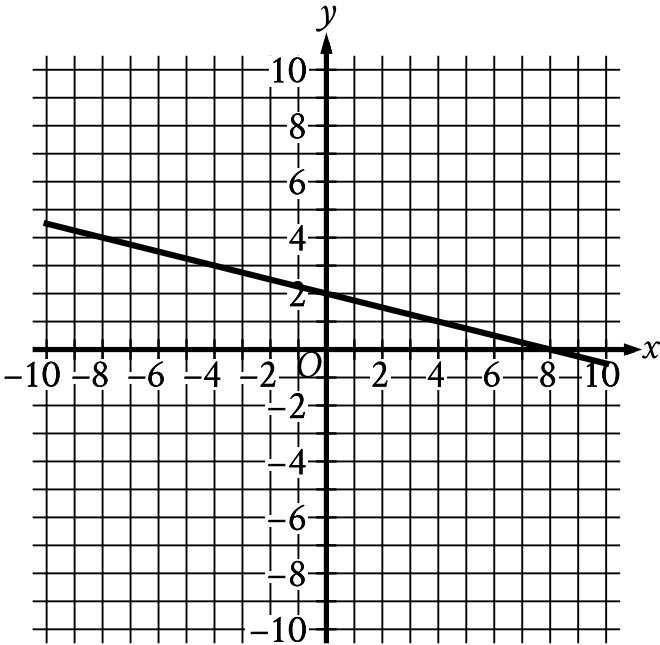
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Data set A consists of the heights of **75** objects and has a mean of **25** meters. Data set B consists of the heights of **50** objects and has a mean of **65** meters. Data set C consists of the heights of the **125** objects from data sets A and B. What is the mean, in meters, of data set C?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question



The graph of $y = f(x) + 14$ is shown. Which equation defines function f ?

- Answer
- A. $f(x) = -\frac{1}{4}x - 12$
 - B. $f(x) = -\frac{1}{4}x + 16$
 - C. $f(x) = -\frac{1}{4}x + 2$
 - D. $f(x) = -\frac{1}{4}x - 14$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$x - y = 1$

$x + y = x^2 - 3$

Which ordered pair is a solution to the system of equations above?

Answer

- A. $(1 + \sqrt{3}, \sqrt{3})$
- B. $(\sqrt{3}, -\sqrt{3})$
- C. $(1 + \sqrt{5}, \sqrt{5})$
- D. $(\sqrt{5}, -1 + \sqrt{5})$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

Sample	Percent in favor	Margin of error
A	52%	4.2%
B	48%	1.6%

The results of two random samples of votes for a proposition are shown above. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

Answer

- A. Sample A had a smaller number of votes that could not be recorded.
- B. Sample A had a higher percent of favorable responses.
- C. Sample A had a larger sample size.
- D. Sample A had a smaller sample size.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

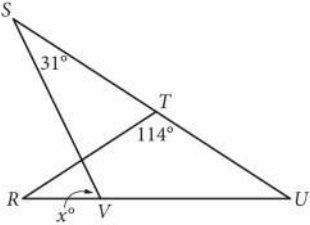
Line ℓ in the xy -plane is perpendicular to the line with equation $x = 2$. What is the slope of line ℓ ?

Answer

- A. 0
- B. $-\frac{1}{2}$
- C. -2
- D. The slope of line ℓ is undefined.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, $RT = TU$. What is the value of x ?

Answer

- A. 72
- B. 66
- C. 64
- D. 58

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$y = x^2 + 3x - 7$

$y - 5x + 8 = 0$

How many solutions are there to the system of equations above?

Answer

- A. There are exactly 4 solutions.
- B. There are exactly 2 solutions.
- C. There is exactly 1 solution.
- D. There are no solutions.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

If $\frac{x+6}{3} = \frac{x+6}{13}$, the value of $x + 6$ is between which of the following pairs of values?

- Answer**
- A. -7 and -3
- B. -2 and 2
- C. 2 and 7
- D. 8 and 13

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by **24%** every **22.96** minutes. Which of the following equations could represent this model, where ***M*** is the estimated mass, in grams, of the sample ***t*** minutes after the lab analyst began observing the sample?

Answer

- A. $M = 100(0.24)^{t+22.96}$
- B. $M = 100(0.24)^{\frac{t}{22.96}}$
- C. $M = 100(0.76)^{t+22.96}$
- D. $M = 100(0.76)^{\frac{t}{22.96}}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$(x - 1)^2 = -4$

How many distinct real solutions does the given equation have?

Answer

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $(3x - 23)(19x + 6)$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. What is the value of b ?

Module 03 • answer sheet

Write one answer per line. SPR: integer, decimal, or fraction.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

15. _____

16. _____

17. _____

18. _____

19. _____

20. _____

21. _____

22. _____